
PROFILE SUMMARY

- Data Scientist with 6 years of experience building production ML and statistical systems across e-commerce search, two-sided marketplaces, and travel personalization, specializing in causal inference, production-grade modeling, and trustworthy experimentation.
- Solid technical background across modeling (PyTorch, XGBoost, scikit-learn), experimentation (A/B Testing, CUPED, Bayesian methods), data tooling (SQL, Pandas, DuckDB), production stacks (Feast, MLflow, Airflow), and cloud (AWS, GCP) with strong scripting fundamentals in Python and SQL.
- Deep expertise in causal inference, uplift modeling, calibration analysis, and fairness auditing, leveraging methodologies such as synthetic control, propensity-score matching, and stratified A/B design to deliver defensible, reproducible, and decision-grade insights.
- Engaged collaborator working cross-functionally with Product, ML Engineering, and Business teams in Agile environments, contributing to roadmap planning, experiment review, and modeling-decision retrospectives with a pragmatic, outcome-oriented mindset.
- Emerging leader who shares technical excellence and fosters a culture of statistical rigor and reproducibility through code reviews and notebooks, while leading applied-science guild sessions and authoring widely adopted experimentation templates.

TECHNICAL SKILLS

Languages & Scripting:	Python, SQL, R, Bash
Modeling Frameworks:	PyTorch, TensorFlow, XGBoost, LightGBM, scikit-learn
Experimentation:	A/B Testing, CUPED, Bayesian methods, Sequential testing
Data Tooling:	Pandas, DuckDB, NumPy, dbt, Spark
MLOps & Production:	Feast, MLflow, Airflow, Kubeflow, BentoML
Visualization & Storytelling:	Streamlit, Plotly, Tableau, Looker, Jupyter
Cloud Platforms:	AWS (S3, SageMaker, EMR), GCP (BigQuery, Vertex AI)
Statistical & Causal Methods:	Hypothesis testing, regression, propensity scoring, causal inference

EDUCATION

Carnegie Mellon University M.S. in Statistics & Machine Learning

Pittsburgh, PA • Sep 2017 - May 2019

WORK EXPERIENCE

Airbnb Data Scientist

New York, NY • Aug 2022 - Present

- Own the recommendation modeling stack across 2 product squads, supporting millions of nightly bookings and 300+ concurrent experiments; lead end-to-end ownership across model design, causal validation, and production reliability within a modern ML platform.
- Designed and shipped a two-tower retrieval model in PyTorch that lifted booking conversion by 12% over the prior gradient-boosted ranking baseline; trained on 2B+ session events on distributed GPU clusters orchestrated by Ray, with feature snapshots backed by Iceberg.
- Designed and analyzed 40+ A/B tests across search ranking, pricing, and onboarding flows using stratified sampling, CUPED variance reduction, and Bayesian sequential testing; caught 3 confounded results before launch and tightened the detection horizon by 35%.

- Built a causal-inference framework on observational booking data using **propensity-score matching**, **difference-in-differences**, and **synthetic control**; reattributed **\$8M** of incremental revenue and informed a marketing-budget reallocation that lifted CAC efficiency by **18%**.
- Partnered with ML Engineering to ship the recommendation model to real-time serving with **Feast** feature integration, **shadow traffic** validation, **Prometheus** drift dashboards, and weekly retraining DAGs on **Airflow**; sustained **99.95%** inference availability and surfaced **11** silent distribution shifts over six quarters.
- Led the exploratory analysis on **pricing fairness** across host segments using **Pandas**, **DuckDB**, and **Plotly**; surfaced **3** systematic biases against new hosts, reframing the modeling roadmap and unlocking a **\$5M** revenue opportunity the team had previously missed.
- Authored **research notes**, **exec readouts**, and **Streamlit dashboards** that translated model behavior for **Product**, **Marketing**, and **Exec leadership**, with **4** papers picked up as the internal reference for the search-ranking surface.

Booking.com **Data Scientist**

Boston, MA • Jul 2019 - Jul 2022

- Built and maintained **200+** production features in the shared feature store feeding **hotel ranking**, **personalization**, and **fraud detection** models, handling **skewed distributions**, **target encoding** for missing data, and time-leakage controls in CV folds; cut feature-engineering time per project by **55%**.
- Established the model-validation rubric for ranking and classification, layering **stratified CV**, **fairness audits** across demographic slices, and **reliability curves** for calibration; caught overfitting in **2** production-ready candidates before launch.
- Designed a **GBM ensemble** for destination-search relevance, applying **SHAP** interpretability, time-based cross-validation, and **Optuna** hyperparameter tuning; lifted **precision@5** by **9%** over the prior heuristic baseline.
- Worked closely with **Product**, **Marketing**, and **Legal** across four markets and three regulatory regimes to translate model outputs into **GDPR-compliant** decisions, shipping shared SQL playbooks and Looker dashboards adopted by **12+** analysts.